BTS–VII–11.17–1250 Reg. No. | | | | | | | | |

A

B.Tech. Degree VII Semester Examination November 2017

ME 1702 VIBRATION AND NOISE CONTROL

(2012 Scheme)

Time : 3 Hours Maximum Marks : 100

PART A

(Answer *ALL* questions)

($8 \times 5 = 40$)

I. (a) Determine the natural frequency of a vibratory system having a mass suspended from the free end of a massless spring. What is the effect of inertia of the spring mass?

(b) Define transmissibility. Explain with the help of transmissibility vs. frequency ratio curves at various damping ratios.

(c) Explain the terms mode shapes and modal matrix.

(d) Determine the ratio of amplitudes of torsional vibrations of a two rotor system.

(e) Derive an expression for the critical speed of a rotating shaft.

(f) Describe Rayleigh's method to find the natural frequency of transverse vibrations of a shaft carrying three point loads.

(g) Differentiate between spherical and cylindrical acoustic wave fronts.

(h) Explain any one method for measuring hearing ability.

PART B

($4 \times 15 = 60$)

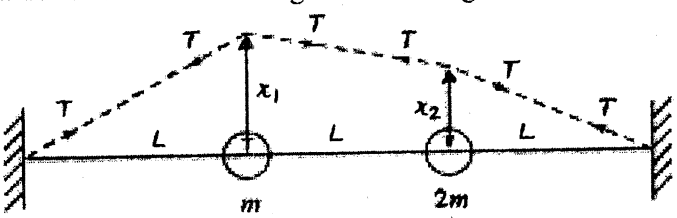
II. (a) Prove that the ratio of successive amplitudes in an under damped vibrating system is a constant. (5)

(b) Determine the time in which the mass in a damped vibrating system would settle down to $1/50^{th}$ of its initial deflection for the following data: m=200 kg, ζ=0.22, k = 40 N/mm. Also find the number of oscillations completed to reach this value of deflection. (10)

OR

III. Determine the frequency of free vibration, when a body of mass 20 kg is suspended from a spring which deflects 15 mm under the weight of the body. Also find the viscous damping required to make the motion critically damped at a speed of 1 m/s. If when damped to this extent, a disturbing force having a maximum value of 187.5 N and vibrating at 8 Hz is made to act on the body, find the amplitude of ultimate motion. (15)

IV. Determine the two natural frequencies and mode shapes for the system shown in figure. The string is stretched with a large tension T and it does not change for small values of oscillation. Length of the string is 3L. (15)



OR

(P.T.O.)

V. (a) What is a torsionally equivalent shaft? Explain. (5)

(b) Determine the natural frequency of torsional vibrations of a shaft with two circular discs of uniform thickness at the ends. The masses of the discs are $M_1 = 500\,kg$ and $M_2 = 1000\,kg$ and their outer diameters are $D_1 = 125\,cm$ and $D_2 = 190\,cm$. The length of the shaft is 300 cm and its diameter is 10 cm. $G = 83\ GN/m^2$. (10)

VI. (a) A thin and uniform rod is subjected to axial force causing longitudinal vibrations. Derive the wave equation. (5)

(b) A uniform beam fixed at one end and free at the other is having transverse vibrations. Derive the frequency equation for the vibrating system. (10)

OR

VII. Determine the frequency of transverse vibration of a 40 mm diameter shaft which is simply supported at the ends and 2.5 m in length. The shaft carries three point loads of masses 30 kg, 70 kg and 45 kg at 0.5 m, 1 m, and 1.7 m respectively from the left support. Neglect the weight of the shaft. $E = 200\ GN/m^2$. (15)

VIII. (a) What is a decibel? Explain addition and averaging of decibels. (6)

(b) Explain the following terms as applied to nose: (9)

(i) Sound Pressure Level (ii) Noise exposure limits

(iii) Psychophysical indices.

OR

IX. (a) Write notes on (i) sound enclosures and (ii) sound barriers used in noise control (5)

(b) Differentiate between temporary and permanent hearing loss from continuous noise. (5)

(c) Explain A, B and C weighting networks. Why are they needed? (5)

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